

Skills Summary

Mixed Decimal Review

Use this reference while completing your worksheet or when studying.

Skill 1 — Round First, Then Estimate

Rule: round each number to the nearest whole number, then compute with the rounded numbers. Use the estimate to check whether an exact answer is reasonable. If both numbers were rounded the same direction, you also know whether the true answer is bigger or smaller than the estimate.

Example: Estimate 5.8×3.2 .

Step 1. An estimate wants easy numbers — round each factor to the nearest whole number before multiplying.

Step 2. $5.8 \rightarrow 6$ and $3.2 \rightarrow 3$, so the product is about 6×3 .
 $\Rightarrow 5.8 \times 3.2 \approx 18$

Try it: Estimate $11.7 \div 3.9$.

check: 3

Skill 2 — Add and Subtract: Line Up the Point

Rule: line up the *decimal points* (never the last digits), fill empty places with zeros, then add or subtract as usual. The point in the answer drops straight down.

Example: Add $4.6 + 3.75$.

Step 1. Line up the decimal points and give both numbers the same number of decimal places before adding.

Step 2. $4.60 + 3.75$: hundredths $0 + 5$, tenths $6 + 7$ regroups, wholes $4 + 3 + 1$.
 $\Rightarrow 4.6 + 3.75 = 8.35$

Try it: Subtract $7 - 1.25$.

check: 5.75

Skill 3 — Multiply, Then Place the Point

Rule: multiply as whole numbers, then give the product as many decimal places as *both factors have together* (tenths $\times$ tenths = hundredths).

Example: Multiply 0.6×0.8 .

Step 1. Multiply the digits as whole numbers, then count decimal places in both factors to place the point.

Step 2. $6 \times 8 = 48$; one place + one place = two places.
 $\Rightarrow 0.6 \times 0.8 = 0.48$

Try it: Multiply 1.5×0.4 .

check: 0.6

Skill 4 — Divide with a Whole-Number Divisor

Rule: shift the decimal point in *both* numbers until the divisor is whole. Both numbers grow tenfold together, so the quotient stays the same.

Example: Divide $4.8 \div 0.6$.

Step 1. Make the divisor a whole number by shifting both numbers one place to the right.

Step 2. $4.8 \div 0.6 = 48 \div 6$

$$\Rightarrow 4.8 \div 0.6 = 8$$

Try it: Divide $6.3 \div 0.7$.

check: 9